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REPORT

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Disparities in patients' perceptions on access to healthcare: implications to promote health equity in Palestine

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ABSTRACT
Equitable access to healthcare is essential for achieving Universal Health Coverage (UHC) and improving health outcomes. This cross-sectional study examines patients' perceived access to healthcare services in the West Bank, providing insights into patients' perceptions of health disparities. Data were collected using stratified convenient sampling of 486 chronic patients at governmental primary healthcare centers in three governorates. A 5-points Likert scale questionnaire was used, based on the '6A' dimensions for healthcare access: affordability, acceptability, accommodation, accessibility, availability, and awareness. The overall healthcare access mean score was 3.29 (SD = 0.46), with only 38.5% of participants reporting satisfactory access levels. Only acceptability (mean =3.75, SD = 0.55), and awareness (mean = 3.66, SD = 0.61) dimensions demonstrated significantly positive perceptions. Bivariate analysis identified income as the main significant determinant for access disparities (Kruskal–Wallis Test χ^2= 75.9, $p < 0.001$). Place of residency, education and income significantly contributed to the disparities within the different access dimensions. The findings highlight significant challenges in healthcare access in the West Bank, particularly regarding services availability and financial affordability. These barriers extremely affect vulnerable populations, exacerbating existing health inequities and undermining efforts toward UHC. The findings emphasise the urgent need for policy interventions to address financial protection, expand service availability, and strengthen healthcare provision.

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Perceived health access; Health disparities; Universal health coverage; Affordability; Accessibility; West Bank; Palestine

Introduction

Equity in health implies ensuring full access to high-quality healthcare services for all individuals and community groups (Whitehead & Dahlgren, 2021). Equitable access to essential healthcare services is fundamental to achieving UHC and enhancing social welfare (WHO, 2014). Literature suggests that inequities and barriers in health access remain prominent in most countries, especially in low-middle-income countries. Equitable access to healthcare is affected by several factors including the availability, accommodation and appropriateness of healthcare services (Gulliford et al., 2002; Jindal et al., 2023; Oliver & Mossialos, 2004).

The concept of healthcare access has been defined differently across health systems reflecting their structural values. Penchansky and Thomas (1981) have defined five dimensions for access,

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including availability, accessibility, affordability, accommodation, and acceptability (Penchansky & Thomas, 1981). The first two dimensions are spatial in nature, availability refers to the quantity and type of services that the citizen can choose to receive, the accessibility dimension refers to the travelling time and the geographic distance between the citizen's residence and the service center (Hoseini-Esfidarjani et al., 2021; Kyriopoulos et al., 2014). The other three dimensions are related to the cost, service quality, and cultural factors. The affordability is related to the financial capacity for individuals to pay for healthcare services. The accommodation dimension is related to how the services are aligned with the population's needs, including working hours and appointment systems. The acceptability dimension is related to the compatibility between patients' cultural values, beliefs, and preferences, and the characteristics of the healthcare providers (Hoseini-Esfidarjani et al., 2021; Shrestha, 2010). Saurman (2016) added the 'awareness' dimension to the model proposed by Penchansky and Thomas, awareness is related to the individuals' recognition of the need for health care services, and their knowledge of the available services and how to access them (Saurman, 2016). Such models have been used to develop tools and questionnaires for assessing access to healthcare from a patient's perspective in several countries (European Patients Forum, 2016; Hoseini-Esfidarjani et al., 2021; Kyriopoulos et al., 2014)

The Palestinian political landscape marked by ongoing Israeli occupation has imposed substantial restrictions on movement between Palestinian cities and villages. These restrictions have created a significant socioeconomic disparity in the Palestinian community. The division of the West Bank under the Oslo Accords, into East Jerusalem and Areas A, B and C, each is governed by different administrative and security regulations, has further complicated access to health care, education, and economic opportunities (World Bank, 2023).

Access to healthcare services in Palestine is shaped by complex contextual and structural dynamics, compounded by the ongoing political conflict and movement restrictions (Giacaman et al., 2009; WHO, 2022). Barriers for accessing healthcare services in Palestine have been widely described in literature as physical barriers caused mainly by the occupying power and dramatically affecting those who are living in remote and rural areas (Jabali et al., 2024; Rijke, 2021). Such delays have dramatically affected the health outcomes of Palestinian patients, causing increased morbidity and mortality specially among individuals with chronic illnesses or cancer. These patients often require timely specialised care that is only available in East Jerusalem hospitals, yet many face significant barriers to access, including prolonged delays or outright denial of permits. As a result, their conditions may deteriorate due to interrupted treatment, delayed diagnosis, or lack of continuity in care (WHO, 2022).

Given the unique challenges in Palestine, including political instability, economic constraints, geographical and social disparities, and restricted mobility, it is essential to employ tools that capture patients' perceived access to healthcare. This study aims to assess chronic patients' perceptions regarding access to healthcare services. Such assessment can provide insights into the barriers and health access inequities from beneficiaries' perspectives that might not be evident from system-level data. It will also highlight areas where interventions are needed to promote equitable healthcare access.

Methods

Study settings

This is a cross-sectional study on patients' perceived access to healthcare services in the West Bank. The study targeted adult Palestinians diagnosed with chronic non-communicable diseases (Cardiovascular & Diabetes Mellitus) who were visiting Ministry of Health (MoH) Primary Health Care (PHC) centers within their localities. Patients were selected from the three largest and the most populated governorates in the West Bank: Nablus in the north, Ramallah in the middle and Hebron in the south of West Bank (MOH, 2023).

The tool

The Perceived Access to Health Care Questionnaire (PAHCQ) was used as the primary tool (Alqutub, 2022; European Patients Forum, 2016; Hoseini-Esfidarjani et al., 2021; Saurman, 2016). The initial questionnaire originally adapted from Alqutub's study was revised to ensure its alignment with the Palestinian context and validated by a selected group of health experts.

The PAHCQ composed of 42 items encompass the six dimensions of healthcare access outlined in Penchansky and Thomas' model and Saurman's study (European patient forum, 2016; Saurman, 2016). The questionnaire is composed of two main sections, the first section includes the demographic and socioeconomic information, the second section includes the six mentioned dimensions: the accessibility dimension (5 items), availability dimension (6 items), acceptability (7 items), accommodation (7 items), awareness (5 items), and affordability (8 items). An additional section about perceived discrimination (2 items) was added to enhance the assessment of perceived health disparities. This addition aligns with similar components used in previous studies, ensuring consistency and comparability with existing research (Alqutub, 2022; European Patients Forum, 2016).

The questionnaire is mainly based on five-point Likert scale questions (scale from 1 to 5), and also includes open ended and multiple-choice questions to ensure comprehensive data collection.

Sampling size and approach

Participating primary healthcare (PHC) facilities were selected using a stratified sampling approach, each of the three regions of the West Bank was represented by a single governorate (the most populated), namely, the northern, middle, and southern regions. The selected governorates were Nablus in the north, Ramallah in the middle and Hebron in the south of West Bank.

The required sample size was calculated using Raosoft® sample size calculator (Raosoft, 2004), resulting in a sample size of 378 patients. The calculation was based on a 95% confidence level, a 5% margin of error and an assumed response rate of 50%, this assumption is commonly used in sample size estimation when the true population proportion is unknown, as it yields the maximum required sample size and ensures a conservative estimate (Raosoft, 2004).

To account for potential non-response and to ensure better representativeness, the investigator increased the sample to 480 individuals, ensuring an equal distribution of 160 participants per governorate. Equal sample sizes maximise statistical power, as power decreases when the sample sizes are unequal or when population differences increase (Rusticus & Lovato, 2019).

Inclusion criteria

Inclusion criteria for this study included adult patients with chronic or long-term conditions, specifically those diagnosed with diabetes and/or cardiovascular diseases for at least one year prior to the study.

Data collection

Data was collected using a self-completed questionnaire strategy, paper copies of the questionnaire were distributed to participants and later processed electronically by data collectors. Six independent data collectors were employed to distribute questionnaires and collect data directly from the patients at PHC centers, they also helped patients who were not able to read or complete questionnaires personally for any reason The data collectors were trained on research ethical considerations including the principles of non-interference and avoiding leading gestures or facial expressions, in addition they were chosen as completely independent data collectors from outside the health system, so as to mitigate any possible data collection or social desirability bias. The data collection process was conducted between May to August 2024.

Validity and reliability

The questionnaire was initially revised by the researchers where some items were modified to ensure cultural and contextual relevance to with the Palestinian settings. It was then further reviewed by seven public health experts from the MoH and academia, and subsequently refined based on their suggestions. All experts confirmed its relevance and validity with the main purpose of the study and its appropriateness within the Palestinian context.

To assess content validity, both Item-Content Validity Index (I-CVI) and Scale-Content Validity Index (S-CVI) were calculated, with both achieving the highest possible value of 1.0.

A pilot test was conducted with a sample of 20 patients to evaluate the reliability of the instrument. Internal consistency was assessed using Cronbach's alpha for both the pilot sample and again after data collection for the whole study sample. Cronbach's alpha coefficients for the whole study sample ranged from 0.74 to 0.90 across the six dimensions, indicating acceptable to excellent reliability. The overall Cronbach's alpha was 0.88 (95% CI: 0.86–0.89), confirming the instrument's reliability and suitability for assessing healthcare access in the study population.

Data analysis

Data analysis was performed using R statistical analysis software (version 4.4.2). Bivariate analysis was conducted to examine the relationships between sociodemographic factors and healthcare access. The Shapiro–Wilk test was used to assess the normality of the mean score, while the Wilcoxon Signed-Rank Test was utilised to determine whether mean scores exceeded the hypothesised threshold of 3.5. The Mann–Whitney U test to evaluate gender differences, the Kruskal–Wallis test to examine differences across the factors of education, place of residence, income, and employment sectors. Spearman's rank-order to assess the associations with age.

A cut-off point of 3.5 on a 5-point Likert scale is often used in research to categorise attitudes, as it represents a logical threshold between negative or neutral attitude and positive agreement. The interpretation of mean scores on the Likert scale is typically as follows: scores between 1.0-2.4 indicate a negative attitude, scores between 2.5-3.4 reflect a neutral attitude, and scores between 3.5-5.0 indicate a positive attitude (Lindner & Lindner, 2024). Therefore, mean scores above 3.5 indicate agreement or positive attitude, while scores below 3.5 suggest neutrality or disagreement.

Results

The study sample comprised 486 respondents from diverse demographic and socioeconomic backgrounds, as detailed in Table 1. The majority of participants were females (62.6%). Over half of the respondents lived in rural localities (56%), more than half were unemployed (54.1%), and 25.7% earned less than the minimum wages of 1,850 NIS. Mean age of the participants was 54.80 (SD = 14.5).

Shapiro–Wilk test was conducted for the collected data at both the overall scores and the dimensional level. Results indicated significant departures from univariate normality. The W statistics for Shapiro–Wilk test ranged from 0.858 (acceptability) to 0.974 (availability and overall access), and the associated p value was below the level of significance, signifying that each distribution deviated significantly from the normal curve.

Although the data did not meet the assumptions of normality, means and standard deviations were reported for descriptive purposes, and clearer comparison. They are preferably used in several research for clearer comparison and more effective and informative analysis in the cases of scale variables and large sample size (Ghasemi & Zahediasl, 2012; Lydersen, 2020), which is applicable to this research paper.

1. Perceived Healthcare Access Results

1.1. Overall Access Score

Table 1. Demographic and socioeconomic characteristics of the study sample.

Demographic characteristic		Number (Percentage)
Gender	Male	182 (37.4%)
	Female	304 (62.6%)
Education	Illiterate	31 (6.4%)
	Elementary	62 (12.8%)
	Intermediate	97 (20.0%)
	Secondary	134 (27.6%)
	University/ Higher education	162 (33.3%)
Place of residency	Camp	20 (4.1%)
	City	194 (39.9%)
	Village	272 (56.0%)
Employment sector	Private and NGOs sector	55 (11.3%)
	Government sector	83 (17.1%)
	Self-employed	51 (10.5%)
	Retired	34 (7.0%)
	Unemployed	263 (54.1%)
Marital status	Single	54 (11.1%)
	Married	362 (74.5%)
	Widow	61 (12.6%)
	Divorced/ separated	9 (1.8%)
Monthly income	Less than 1850 NIS	125 (25.7%)
	1850–3000 NIS	183 (37.7%)
	3001–5000 NIS	138 (28.4%)
	More than 5000 NIS	40 (8.2%)
Age	Middle-aged (less than 50 years)	152 (31%)
	Pre-retirement (51–60)	151 (31.5%)
	Older adults (61+)	183 (37.5%)

The analysis revealed significant concern regarding healthcare access in the West Bank. Using a cut-off point of 3.5 to indicate minimum positive perceptions, the overall healthcare access mean score was 3.29 (SD = 0.46), with only 38.5% of participants reporting satisfactory access levels (scores $\geq$ 3.5). The One-Sample Wilcoxon Signed-Rank Test (W = 31,837.0, p = 0.999) indicates a non-statistically significant results, suggesting insufficient evidence to conclude that perceptions were significantly above the acceptable threshold.

Among the six dimensions, only acceptability (mean =3.75, SD = 0.55) and awareness (mean = 3.66, SD = 0.61) dimensions demonstrated significantly positive perceptions exceeding the hypothesised threshold of 3.5, while the remaining dimensions did not reach adequate levels as shown in Table 2 below.

1.2. Accessibility to Healthcare Services

More than half of participants (55.6%) reported positive perceptions of accessibility (scores $\geq$ 3.5) with a mean score of 3.35 (SD = 0.71). However, the One-Sample Wilcoxon Signed-Rank

Table 2. Perceptions of health care access by its dimensions.

Dimension	Agreement percentage	Mean (SD)	Wilcoxon Signed-Rank Test (Statistic P-Value)	The confidence interval (95%)
Acceptability	75.9%	3.75 (0.55)	92983.0 (< 0.001)	[3.79, ∞)
Awareness	73.0%	3.66 (0.61)	85026.5 (< 0.001)	[3.70, ∞)
Accessibility	55.6%	3.35 (0.71)	52834.5 (0.980)	[3.40, ∞)
Availability	31.3%	3.19 (0.64)	26520.0 (0.999)	[3.08, ∞)
Accommodation	30.0%	3.17 (0.69)	31618.0 (0.999)	[3.14, ∞)
Affordability	26.7%	2.60 (1.24)	17885.5 (0.999)	[2.43, ∞)
Overall access score	38.5%	3.29 (0.46)	31837.0 (0.999)	[3.27, ∞)

Note. The confidence interval (CI) is one-sided (lower bound) due to the Wilcoxon signed-rank test's alternative hypothesis (scores $\geq$ 3.5).

Test (W = 52,834.5, p = 0.980) indicated that these perceptions were not statistically significant. While most participants expressed positive perceptions regarding the physical distance and travel time with 72.6% agreeing that it is easy to physically access their regular healthcare center (M = 3.59, SD = 0.95). Similarly, 74.5% agreed that the distance between their home and the healthcare center is appropriate (M = 3.58, SD = 0.96), and 75.5% felt that the time needed to reach the healthcare center is appropriate (M = 3.61, SD = 0.96).

Over half (57.8%) reported their need to visit another center or city to obtain the required healthcare services (M = 2.86, SD = 1.18), and only 47.7% agreed that all the services they need are available at their regular healthcare center (M = 3.13, SD = 1.02).

1.3. Availability of Healthcare Services

Only 31.3% of participants reported positive perceptions for the availability dimension, with a mean score of 3.19 (SD = 0.64). The One-Sample Wilcoxon Signed-Rank Test (W = 26,520.0, p = 0.999) confirmed that these perceptions were not statistically significant. The lowest agreement was found for the availability of medications and laboratory tests. Access to free medications and routine examinations was limited, with only 26.7% agreeing they could obtain necessary medications monthly (M = 2.60, SD = 1.00) and 30.2% stating they had free access to routine health monitoring exams (M = 2.69, SD = 1.04). Similarly, only 42.8% felt that all necessary healthcare services were available at their regular center (M = 3.03, SD = 1.01), and 47.3% agreed that specialist and consultant services were accessible as needed (M = 3.12, SD = 1.01).

1.4. Acceptability

The acceptability dimension had the highest positive agreement rates, with 75.9% of participants providing agreement scores (M = 3.75, SD = 0.55). The One-Sample Wilcoxon Signed-Rank Test (W = 92,983.0, p < 0.001) confirmed statistically significant positive perceptions. All statements within acceptability dimension had mean scores above 3.5 and relatively high agreement rates, indicating uniformly positive perceptions across all aspects of acceptability and quality. The highest agreement (89.1%) was reported for statements regarding respectful treatment, confidentiality, cultural sensitivity, and appropriate communication by the medical team.

1.5. Accommodation

Only 30% of participants showed positive agreement for the accommodation dimension. The overall mean score was also low (M = 3.17, SD = 0.69), and One-Sample Wilcoxon Signed-Rank Test (W = 31,618.0, p = 0.999) demonstrated significantly lower perceptions compared to the threshold value of 3.5. The lowest agreement was found for the statement regarding considering patient feedback about the quality of healthcare by medical team (M = 2.85, SD = 1.01, 32.5% agreement), and the availability of assistive tools for people with disabilities and the elderly (M = 3.01, SD = 1.02, 37.9% agreement). Perceptions of waiting times (M = 2.90, SD = 1.18, 45.7% agreement) and communication with medical teams via phone or social media (M = 2.92, SD = 1.16, 43.2% agreement) were also relatively low. Conversely, higher agreement was found for working hours (M = 3.44, SD = 0.97, 67.3%) and the healthcare center's environment (M = 3.64, SD = 0.96, 72.2%).

1.6. Awareness

73% of participants reported positive agreement with the awareness dimension. The mean score for all participants was also relatively high (M = 3.66, SD = 0.61), and the One-Sample Wilcoxon Signed-Rank Test (W = 85,026.5, p < 0.001) confirmed statistically significant positive perceptions of this dimension. The highest scores were observed for participants' understanding of educational

messages (M = 3.82, SD = 0.70, 81.9% agreement), communication with the medical team (M = 3.78, SD = 0.68, 81.9% agreement), and the use of simple and appropriate language (M = 3.79, SD = 0.76, 80.9% agreement). In contrast, lower agreement rates were reported regarding instructions on healthy lifestyle habits (M = 3.35, SD = 1.06, 61.5% agreement).

1.7. Affordability

The affordability dimension emerged as the most critical barrier to healthcare access, with a low mean score of 2.60 (SD = 1.24), and only 26.7% of participants reporting adequate financial access. The One-Sample Wilcoxon Signed-Rank Test (W = 31,837.0, p = 0.999) confirmed that affordability perceptions were significantly below the hypothesised threshold. The results showed that the greatest financial barriers were paying for critical and intensive care services (M = 1.95, SD = 1.70, 23.7% agreement) and psychologist clinics (M = 2.06, SD = 1.77, 26.5% agreement). Other major challenges included costs associated with hospital stays (M = 2.19, SD = 1.83, 29.8%), assistive medical devices (M = 2.29, SD = 1.87, 32.3%), and physical therapy services (M = 2.40, SD = 1.91, 35.0%).

Additionally, 53.5% of participants reported their need of additional financial support for medical treatment (M = 3.14, SD = 2.00), while 56% had to cut spending on essentials or borrow money to afford healthcare (M = 3.24, SD = 1.99).

1.8. Perceived Discrimination

When asked about experiences of discrimination or stigma in healthcare, 63 participants (12.9% of the total) reported perceiving discrimination. The most commonly cited reason was low income (60% of cases), followed by illness or disability (25% of cases).

2. Demographic Disparities in Healthcare Access

A bivariate analysis was used to examine whether perceptions of healthcare access varied based on demographic variables among Palestinians in the West Bank. The detailed findings of this analysis are shown in Table 3, and the concise results of the statistical testing (Mann–Whitney U Test and Kruskal–Wallis Test) for the variations in different dimensions due to demographic factors are summarised in Table 4. Income was the only significant determinant for the overall perceived healthcare access scores (Kruskal–Wallis Test χ^2= 75.9, p value =0.000). However, other demographic factors significantly contributed to the disparities within the different health access dimensions.

Educational level was a significant factor in differences across the accessibility dimension (Kruskal–Wallis χ^2 = 12.1, p = 0.016) and affordability dimension (χ^2= 48.09, p value =0.000), participants with higher educational levels reported better accessibility and affordability. Place of residence contributed to disparities in multiple dimensions: accessibility (χ^2 = 22.6, p < 0.001), awareness (χ^2 = 10.6, p = 0.005), affordability (χ^2 = 10.7, p = 0.005), and accommodation (χ^2 = 25.5, p < 0.001). Residents of camps reported the highest mean scores for accessibility, accommodation and awareness, while villages' residents reported the lowest mean score of accessibility and awareness. City residents reported the lowest mean score of accommodation and the highest score of affordability. There were no significant differences in overall access score between males and females (Mann–Whitney U = 25425.5, p = 0.135), nor across different educational levels (χ^2 = 6.2, p = 0.185).

Better results seen in refugee camps that are usually served by UNRWA healthcare services, may be attributed to the comprehensive primary healthcare services provided by UNRWA. This includes free monthly medications, laboratory tests, and health counselling. Additionally, the presence of primary healthcare centers within the camps ensures easy access for all residents. In contrast, individuals living in villages face the greatest challenges due to movement restrictions and access barriers, as specialised and hospital care is mostly concentrated in urban areas (Al Eker et al., 2025).

Table 3. Disparities in healthcare access dimensions by demographic characteristics.

Demographic characteristic	Components	Accessibility score		Availability score		Acceptability score		Accommodation		Awareness score		Affordability score		Overall access score	
		Mean (SD)	(P-Value)	Mean (SD)	(P-Value)	Mean (SD)	Statistic (P-Value)	Mean (SD)	Statistic (P-Value)	Mean (SD)	(P-Value)	Mean (SD)	(P-Value)	Mean (SD)	(P-Value)
Gender	Male	3.29 (0.74)	25055.5 (0.078)[a]	3.18 (0.63)	27394.5 (0.857)[a]	3.80 (0.47)	28825.0 (0.421)[a]	3.15 (0.71)	27070.0 (0.690)[a]	28836.5 (0.420)[a]	28836.5 (0.420)[a]	2.37 (1.18)	22886.0 (0.001)[a]	3.25 (0.43)	25425.5 (0.135)[a]
	Female	3.39 (0.70)		3.19 (0.65)		3.73 (0.59)		3.18 (0.68)		3.64 (0.65)		2.73 (1.25)		3.31 (0.47)	
Education	Illiterate	3.23 (0.72)	12.1 (0.016)[b]	3.24 (0.64)	3.6 (0.458)[b]	3.69 (0.44)	6.7 (0.151)[b]	3.01 (0.66)	8.1 (0.089)[b]	6.4 (0.170)[b]	6.4 (0.170)[b]	2.68 (1.35)	48.09 (0.001)[b]	3.25 (0.49)	6.2 (0.185)[b]
	Elementary	3.16 (0.72)		3.15 (0.60)		3.76 (0.47)		3.23 (0.69)		3.62 (0.54)		2.20 (1.26)		3.19 (0.41)	
	Intermediate	3.29 (0.69)		3.26 (0.70)		3.81 (0.56)		3.25 (0.72)		3.75 (0.49)		2.08 (1.11)		3.24 (0.43)	
	Secondary	3.45 (0.75)		3.23 (0.61)		3.81 (0.48)		3.11 (0.65)		3.73 (0.59)		2.66 (1.21)		3.33 (0.43)	
	University education	3.41 (0.69)		3.12 (0.64)		3.68 (0.64)		3.17 (0.71)		3.58 (0.74)		2.99 (1.16)		3.33 (0.49)	
Place of Residence Type	Camp	3.70 (0.57)	22.6 (0.001)[b]	3.39 (0.64)	2.1 (0.349)[b]	3.74 (0.59)	3.1 (0.212)[b]	3.32 (0.43)	25.5 (0.001)[b]	10.6 (0.005)[b]	10.6 (0.005)[b]	2.59 (1.03)	10.7 (0.005)[b]	3.42 (0.39)	0.8 (0.667)[b]
	City	3.51 (0.65)		3.16 (0.67)		3.72 (0.58)		2.94 (0.79)		3.54 (0.72)		2.82 (1.27)		3.28 (0.50)	
	Village	3.22 (0.73)		3.19 (0.62)		3.78 (0.52)		3.32 (0.58)		3.75 (0.52)		2.44 (1.21)		3.28 (0.43)	
Monthly Income	Less than 1850 NIS	3.04 (0.75)	36.0 (0.001)[b]	2.93 (0.58)	41.6 (0.001)[b]	3.64 (0.62)	17.7 (0.001)[b]	2.90 (0.67)	52.7 (0.001)[b]	31.4 (0.000)[b]	31.4 (0.001)[b]	2.14 (1.15)	66.0 (0.001)[b]	3.03 (0.45)	75.9 (0.001)[b]
	1850 – 3000 NIS	3.41 (0.72)		3.19 (0.61)		3.77 (0.56)		3.15 (0.66)		3.70 (0.65)		2.57 (1.10)		3.30 (0.44)	
	3001 – 5000 NIS	3.50 (0.59)		3.40 (0.65)		3.86 (0.47)		3.44 (0.66)		3.76 (0.58)		2.63 (1.24)		3.43 (0.38)	
	More than 5000 NIS	3.60 (0.64)		3.25 (0.67)		3.67 (0.43)		3.14 (0.67)		3.64 (0.55)		4.02 (1.02)		3.55 (0.43)	
Employment Sector	Private sector	3.27 (0.82)	8.4 (0.134)[b]	3.17 (0.62)	20.5 (0.001)[b]	3.86 (0.41)	3.7 (0.598)[b]	3.08 (0.72)	23.6 (0.001)[b]	16.6 (0.005)[b]	16.6 (0.005)[b]	2.67 (1.29)	16.9 (0.005)[b]	3.29 (0.48)	4.5 (0.480)[b]
	Government sector	3.47 (0.68)		3.41 (0.63)		3.75 (0.63)		3.46 (0.61)		3.76 (0.69)		2.18 (1.01)		3.34 (0.46)	
	NGOs	3.32 (1.32)		3.37 (0.43)		3.34 (0.73)		2.86 (0.80)		3.36 (0.43)		2.49 (1.54)		3.12 (0.65)	
	Self-employed														

		3.42		3.08		3.73		2.99		3.67		2.81		3.28	
		(0.60)		(0.53)		(0.47)		(0.72)		(0.63)		(1.13)		(0.41)	
	Retired	3.55		3.05		3.76		3.13		3.34		3.11		3.32	
		(0.64)		(0.73)		(0.52)		(0.67)		(0.82)		(1.12)		(0.41)	
	Unemployed	3.30		3.16		3.75		3.14		3.67		2.61		3.27	
		(0.71)		(0.65)		(0.56)		(0.69)		(0.55)		(1.29)		(0.46)	
Age	(≤50 years	3.36	0.77	3.28	4.38	3.79	3.95	3.22	2.15	3.77	8.25	2.50	7.00	3.32	2.39
		(0.75)	(0.856)[b]	(0.61)	(0.223)[b]	(0.54)	(0.267)[b]	(0.68)	(0.543)[b]	(0.61)	(0.041)[b]	(1.10)	(0.072)[b]	(0.45)	(0.495)[b]
	(51–60)	3.29		3.21		3.78		3.18		3.69		2.54		3.28	
		(0.78)		(0.63)		(0.55)		(0.70)		(0.65)		(1.22)		(0.48)	
	(≥ 61)	3.39		3.14		3.77		3.13		3.60		2.56		3.26	
		(0.65)		(0.60)		(0.51)		(0.70)		(0.58)		(1.31)		(0.42)	

[a]Statistical Significance of Mann-Whitney U Test.
[b]Statistical Significance of Kruskal-Wallis Test.

Table 4. The statistical tests results and *P*-values for the disparities in perceived healthcare access dimensions by demographic characteristics.

	Accessibility	Availability	Acceptability	Accommodation	Awareness	Affordability	Overall access
Gender	25055.5 (0.078)[a]	27394.5 (0.857)[a]	28825.0 (0.421)[a]	27070.0 (0.690)[a]	28836.5 (0.420)[a]	22886.0 (0.001)[a]	25425.5 (0.135)[a]
Education	12.1 (0.016)[b]	3.6 (0.458)[b]	6.7 (0.151)[b]	8.1 (0.089)[b]	6.4 (0.170)[b]	48.09 (0.001)[b]	6.2 (0.185)[b]
Place of Residence	22.6 (0.001)[b]	2.1 (0.349)[b]	3.1 (0.212)[b]	25.5 (0.001)[b]	10.6 (0.005)[b]	10.7 (0.005)[b]	0.8 (0.667)[b]
Monthly Income	36.0 (0.001)[b]	41.6 (0.001)[b]	17.7 (0.001)[b]	52.7 (0.001)[b]	31.4 (0.001)[b]	66.0 (0.001)[b]	75.9 (0.001)[b]
Employment sector	8.4 (0.134)[b]	20.5 (0.001)	3.7 (0.598)[b]	23.6 (0.001)[b]	16.6 (0.005)[b]	16.9 (0.005)[b]	4.5 (0.480)[b]
Age	0.77 (0.856)[b]	4.38 (0.223)[b]	3.95 (0.267)[b]	2.15 (0.543)[b]	8.25 (0.041)[b]	7.00 (0.072)[b]	2.39 (0.495)[b]

[a]Statistical Significance of Mann-Whitney U Test.
[b]Statistical Significance of Kruskal-Wallis Test.

The overall access score differed significantly across income levels ($\chi^2 = 75.9$, $p < 0.001$). Participants earning less than 1,850 NIS reported the lowest overall access score, with a mean of 3.03 (SD = 0.45), while participants earning more than 5,000 NIS reported the highest overall access score, with a mean of 3.55 (SD = 0.43). Significant differences were also observed across all individual components, with the lowest-income participants reporting the lowest scores. These findings indicate a strong positive relationship between income and perceptions of healthcare access across all dimensions.

Employment sectors contributed to statistically significant differences in the scores for the availability dimension ($\chi^2 = 20.5$, $p = 0.001$), accommodation ($\chi^2 = 23.6$, $p < 0.001$), affordability ($\chi^2 = 16.9$, $p = 0.005$) and awareness ($\chi^2 = 16.6$, $p = 0.005$). Government sector employees reported the highest mean scores across all dimensions except affordability. The overall access score did not show a significant correlation with age. Across the different age categories; the Kruskal–Wallis test indicated no statistically significant differences in the overall access score ($\chi^2 = 2.39$, $p = 0.495$); although younger adults (M = 3.32, SD = 0.45) reported slightly higher overall access scores compared to older adults (M = 3.26, SD = 0.42), the findings suggest that while age does not significantly influence the overall access score, older individuals may face greater challenges related healthcare availability and awareness.

To control for multiple comparisons and reduce the risk of Type I error, adjusted p-values were computed using the Benjamini–Hochberg (BH) procedure. Adjusted p-values were reported for each test but didn't contribute to any changes in the statistical significance results, none of the six component scores reached significance after Benjamini–Hochberg adjustment. Adjusted p-values can be found in the detailed tables in the supplementary files.

Discussion

The study aimed to assess and analyze the patients' perceptions regarding healthcare access and healthcare disparities in Palestine, the results revealed moderately poor perceptions for healthcare access among patients with chronic diseases in the West Bank. The overall perceived access score was below the threshold value of positive access of 3.5, with approximately 61.5% of participants reporting either neutral or negative perceptions. Only two dimensions of health access that demonstrated significantly positive perceptions are acceptability and awareness with 75.9% and 73% rates respectively. Several studies have confirmed relatively good health awareness and knowledge levels among Palestinian patients with chronic diseases (Hemadneh et al., 2019; Sa'adeh et al., 2018;), which has been also linked with good impact on medical adherence and management (Jamous et al., 2014; Najjar et al., 2015). Saádeh and his colleagues observed high knowledge, attitude,

and practice scores of hypertensive patients towards prevention and early detection of chronic kidney diseases (Sa'adeh et al., 2018).

Physical accessibility dimension reported a relatively moderate agreement rate of 55.6% among participants, while service availability dimension showed notably low scores. Physical healthcare access in Palestine has been considered a critical concern, frequently hindered by Israeli-imposed restrictions of checkpoints and imposed movement restrictions in between rural areas in the West Bank (Marie & Battat, 2021; Rijke, 2021; Vitullo et al., 2012). Palestinians in the West Bank are facing hundreds of physical obstacles that permanently or intermittently restrict their movement in the West Bank, including East Jerusalem. The United Nations Office for the Coordination of Humanitarian Affairs (OCHA) has documented about 790 movement obstacles in 2024, which include permanent and partial checkpoints, earth mounds, road gates or roadblocks (OCHA, 2024). Patients requiring specialised care in East Jerusalem hospitals face significant barriers due to Israeli permission requirements which usually don't exceed 75% approval rates in the best conditions (WHO, 2022).

High percentages of participants reported shortages in services within their residency area. Specifically, 73.3% reported their unmet need for medications, while 69.8% indicated a lack of laboratory services. Furthermore, 57.8% of participants reported their need to travel to other centers or cities to obtain necessary services. Chronic shortages of medications and medical supplies frequently highlighted reported in MoH reports and other studies (Marie & Battat, 2021; Mitwalli et al., 2023; MOH, 2024). Medical goods remain the largest driver of household out-of-pocket healthcare expenditures in Palestine (PCBS & MOH, 2025). Shortages of medicines and medical supplies in MoH clinics has become worse during the recent years due to the governmental fiscal crises and the resulting in substantial budgetary shortfalls and increasing arrears owed to local pharmaceutical suppliers (Jadou, 2024). Addressing these systemic financial constraints requires coordinated efforts to improve funding mechanisms, strengthen the supply chain, and ensure equitable healthcare access for all Palestinians.

The accommodation dimension also received poor ratings, reflecting participants' dissatisfaction regarding services sufficiency, long waiting times, medical staff communication and attitudes, and deficiency of assistive tools for elderly or people with disabilities. These findings underline the need to enhance the efficiency of healthcare services to meet patients' needs and expectations, improve resources allocation, and address systemic barriers to comprehensive service delivery.

The affordability dimension recorded the lowest positive agreement rate among participants, reflecting financial accessibility as the most perceived determinant and barrier to healthcare access. This finding aligns with other studies that confirmed financial challenges associated with healthcare access in Palestine (El Jabari et al., 2024; Nasr et al., 2021), Naser et al. confirmed that poorer and less educated people fare worse both in terms of access to health services as well as in health status. Data from Palestinian health accounts highlighted high household out-of-pocket expenditure as a percentage of total health expenditure, which has reached to 38.4% (PCBS & MOH, 2025). Such relatively high rate of household out of pocket expenditure reflects inadequacy in public health services and/or inability to access available services due to physical barriers (WHO, 2022). Furthermore, it underlines inequities within the healthcare system (Rizk & Abou-Ali, 2016), different research papers have linked between non-communicable diseases (NCDs) with elevated out-of-pocket expenses (Fong, 2019; Islam et al., 2014; Park et al., 2021), often leading to catastrophic health expenditures in impoverished settings (M. Ashour, 2017; Ashour, 2023; Rashad & Sharaf, 2015). In Palestine, population growth and the rising prevalence of chronic diseases are considered as the key contributors to increasing healthcare costs (Morrar et al., 2021). Additionally, high poverty and unemployment rates in Palestine are alarming for increased financial barriers for healthcare access (Ashour, 2023), these risks have been amplified by the economic disruption caused by Covid-19 pandemic and the sever Israeli aggression since October 7th, 2023. Poverty in Palestine is projected to rise to 74.3% in 2024 while unemployment rate has raised to 35% in the West Bank (ESCWA & UNDP, 2024).

The study has identified several key factors contributing to disparities in perceived health access, monthly income emerged as the major determinant for disparities in perceived healthcare access, influencing both overall access scores and the scores across six specific dimensions. Additionally, other factors were found to contribute to disparities in some dimensions of healthcare access, including place of residency, education level, age and employment sector. Factors contributing to healthcare access disparities in Palestine can be categorised as economic factors, geographical barrier, social and cultural factors. Economic factors are globally evident to prevent low-income individuals from seeking care. Peters et al. indicates that individuals in poorer nations have significantly less access to health services compared to those in wealthier countries. Moreover, within these nations, poor populations face substantial barriers to obtaining necessary healthcare services (Peters et al., 2008). Economic factors include both direct costs caused by direct payment to medical products or medical services, and indirect costs of healthcare services such as transportation and lost wages.

Geographical barriers mainly affect people living in rural and remote areas that lack adequate healthcare facilities, making health access challenging for their residents. Palestinian case is facing the double burden of distance challenge for remote areas and geopolitical physical access barriers caused by Israeli occupation mentioned above (OCHA, 2024; WHO, 2022). Globally, the literature implies that physical access is considered a major health access barrier in many low- and middle-income countries, distant and inequitable provision of healthcare in rural communities' limit women's willingness and ability to seek healthcare, particularly when appropriate transportation is limited, communications are difficult, and physical or climate conditions are inhospitable (Timyan et al., 2018)

Cultural and social factors include residential disparities, educational level, occupation, cultural beliefs, social stigmas, and lack of awareness can prevent individuals from utilising available healthcare services. Socioeconomic have been evidenced to be as critical determinants in healthcare access disparities and health seeking behaviour within the Palestinian community (Albrithen et al., 2024; Hammoudeh et al., 2017; Majaj et al., 2013). Research evidence have confirmed that health seeking behaviour for Palestinian women is highly determined by their gendered socio-cultural role; their health beliefs; financial affordability and geographical accessibility (Bates et al., 2017; Majaj et al., 2013).

Perceived discrimination reported by some participants are aligned with the findings reported in annual Palestinian Coalition for Accountability and Integrity (AMAN) reports. The results of AMAN public opinion poll show that 44% of citizens who received public service in 2022 felt that they were compelled to use wasta to obtain services, and their feelings were mostly attributed to their personal beliefs and attitudes (AMAN, 2022). In another study that correlated between personal connections or 'wasta' and health access disparities in the Palestinian community, where people are using personal influences to gain unfair advantages for accessing healthcare services or privileges. Wasta is considered a major aspect of corruption and poor governance of the system, and also can be related to tribal and cultural beliefs (Takruri et al., 2023).

Implications to promote equitable healthcare access

When participants were asked for their suggestions for improving healthcare access, two major themes emerged: increasing service availability, and promoting accessibility. Specific recommendations proposed by participants included improving availability of medications and laboratory tests, increasing the number of primary healthcare centers closer to their residential areas, and hiring more healthcare staff, particularly specialised doctors for diabetes and chronic diseases. Participants also emphasised the need to improve patient-staff communication, both in person and telecommunication, provide transportation support, offer psychosocial support for elderly patients with chronic conditions, and extend healthcare facility working hours. These recommendations should be given serious consideration and further analyzed by decision-makers to address the identified disparities and enhance healthcare access effectively.

The findings call for urgent and targeted policy interventions to reduce financial barriers and improve service availability across all localities. Addressing these challenges is essential for

promoting a more inclusive and accessible healthcare system that meets the diverse needs of all community groups in the West Bank. Policymakers must prioritise strategies that align with the principles of equity and inclusivity to ensure comprehensive healthcare access.

Achieving universal health coverage (UHC) and promoting health equity in Palestine requires intensified coordinated efforts to provide essential health services to all individuals, aligned with a holistic approach to improve socioeconomic conditions and social determinants of health in Palestine. Key actions include strengthening and expanding healthcare infrastructure to reduce geographic disparities, expanding the health coverage for chronic patients including essential medinas and healthcare services, promoting financial protection mechanisms to alleviate the burden of healthcare costs, and fostering inclusive policies that address the unique needs of marginalised communities.

In addition, capacity-building initiatives should focus on enhancing healthcare workforce distribution particularly in under-served areas, and ensuring the availability and sustainability of critical resources, such as medications and diagnostic services, promoting sustainable and adequate health financing for the Palestinian health sector. Collaboration between governmental agencies, non-governmental organisations, and international partners is essential to mobilise resources and sustain progress toward equitable and accessible healthcare.

Conclusion

In conclusion, this study underscores the varied perceptions of healthcare access across the West Bank, highlighting significant challenges, particularly in the dimensions of perceived availability and affordability. While the findings reveal strengths in cultural acceptability and communication, disparities in financial accessibility and service provision hinder equitable healthcare access. These barriers disproportionately affect vulnerable populations, exacerbating existing health inequities and undermining efforts toward achieving inclusive healthcare.

The research findings emphasised the need to enhance healthcare structure and delivery approach to achieve better equity and bridging gaps in service provision and health access. Several successful and recommended approaches can be employed to enhance equal health accessibility and expand access among marginalised communities including mobile clinics for Bedouin and remote communities, extending working hours in rural health centers, strengthening primary healthcare provision through family health practice approach, expanding hospital services and resources. Such interventions are crucial to advancing universal health coverage and achieving equitable access to care.

On the longer-term aspects, reforming health insurance in Palestine is also recommended to ensure equitable access and financial protection for all Palestinians (Marmot et al., 2021; WHO, 2018). The reformed insurance scheme should offer comprehensive and affordable coverage, prioritise vulnerable and marginalised groups, and guarantee universal coverage.

By addressing these pressing challenges of healthcare access, service provision and financial protection, stakeholders can work collectively toward a more resilient healthcare system, ultimately realising the vision of UHC in Palestine and promoting the well-being of all its citizens.

Limitations:

Data collected was through self-reported measures, which are subject to recall bias from patients, especially elderly patients.

- The sample considered in the study includes chronic patients of diabetes or cardiovascular diseases, so it may not be fully representative of all social groups or patients' categories.
- To mitigate possible bias, independent data collectors were chosen and trained on ethical research considerations.
- To control for multiple comparisons and reduce the risk of Type I error, adjusted p-values were computed using the Benjamini–Hochberg (BH) procedure. Both original and adjusted p-values are reported for each test and can be found in the detailed tables in the supplementary files.

Ethical approval statement

Ethical approval was obtained from the Research Ethics Committee of Al-Quds University (REF number: 05/ 2023). Official approval from the Palestinian Ministry of Health to conduct the study within primary health care centers was obtained in April 2024.

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Disclosure statement

No potential conflict of interest was reported by the author(s).

Data availability statement

The dataset analyzed during the current study is available from the corresponding author on reasonable request.

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